

A Polygraph popularly referred to as a lie detector, measures and records several physiological indices such as blood pressure, pulse, respiration, and skin conductivity while the subject is asked and answers a series of questions. A Pyrometer is a type of remote-sensing thermometer used to measure the high temperature of a surface. A Gyroscope is a spinning wheel or disc in which the axis of rotation is free to assume any orientation. Kymograph is an instrument for recording variations in pressure, as of the blood or in tension, as of a muscle, by means of a pen or stylus that marks a rotating drum.

52. The instrument, used for measuring angular distances in vertical plane (elevation) and the horizontal plane (azimuth), is :

- (a) Bevel protractor (b) Altimeter
(c) Syncline (d) Theodolite

Chhattisgarh P.C.S. (Pre) 2020

Ans. (d)

Theodolite is an optical instrument, used for measuring angular distances between designated visible points in the horizontal and vertical planes.

53. Consider the following actions :

1. Detection of car crash/collision which results in the deployment of airbags almost instantaneously
2. Detection of accidental free fall of a laptop towards the ground which results in the immediate turning off of the hard drive
3. Detection of the tilt of the smart-phone which results in the rotation of display between portrait and landscape mode

In how many of the above actions is the function of accelerometer required?

- (a) Only one (b) Only two
(c) All three (d) None

I.A.S. (Pre) 2023

Ans. (c)

An accelerometer is a device that measures the vibration, or acceleration of motion, of a structure. One of the most common uses for MEMS accelerometers is in airbag deployment systems for modern automobiles. In this case, the accelerometers are used to detect the rapid negative acceleration of the vehicle to determine when a collision has occurred and the severity of the collision. A free-fall sensor (FFS) is an accelerometer used to detect if a system has been dropped and is falling. It can then apply safety measures such as turning off and parking the head of a hard drive to prevent a head crash and resulting data loss upon impact. Smartphones, digital audio players and personal digital assistants contain accelerometers for user interface control; often the accelerometer is used to present landscape or portrait views of the device's screen, based on the way the device is being held. Hence, in all three given actions, the function of accelerometer is required.

Mechanics

Mass, Motion, Force, Acceleration, Momentum, Work, Power, Energy

Notes

- **Mechanics** : Mechanics is that area of science concerned with the behaviour of physical bodies when subjected to force or displacements and the subsequent effects of the bodies on their environment.
- **Weight** : Weight is the force exerted in a body by gravity.
- **Mass** : Mass represents the amount of matter in a particle or object. The mass of matter is constant at every place. It is denoted by 'm'.

Difference between Weight and Mass

S.N.	Weight (W)	Mass (m)
1.	Gravitational force of Earth	A measurement of the amount of matter something contains
2.	It is a vector quantity.	It is a scalar quantity.
3.	Weight varies according to places.	It is constant at every place.
4.	Its unit is newton.	Its unit is gram or kilogram.

Distance & Displacement :

- In daily language Distance & Displacement are used in the same sense but in Physics both have a different meaning.
- The length of a specific path travelled between two points by an object is distance.
- A displacement is a vector whose length is the shortest distance between the initial to the final position of a point.
- Travelled distance has only magnitude, while displacement has magnitude along with the direction. It is clear that distance is a scalar quantity and displacement is a vector quantity. The travelled distance by an object is not zero, but the final displacement may be zero. Displacement of an object will be zero when the moving object comes to its initial point.

Speed & Velocity :

- Speed is the distance travelled per unit of time.
So, speed = $\frac{\text{travelled distance}}{\text{time}}$
It is a scalar quantity and its SI unit is metre/second.
- Velocity is the distance travelled by an object per unit time in a particular direction.

$$\backslash \quad \text{Velocity} = \frac{\text{Displacement}}{\text{time}}$$

- The difference between speed & velocity is that speed has the only magnitude without any specific direction, while velocity has both magnitude & direction. Speed is scalar & velocity is a vector quantity.

Acceleration :

- Acceleration is the rate of change of velocity of an object with respect of time.

$$\backslash \quad \text{Acceleration} = \frac{\text{rate of change of velocity}}{\text{time}}$$

$$\square \quad \text{Acceleration} = \frac{\text{Final velocity} - \text{initial velocity}}{\text{time}}$$

If the initial velocity of object is u & after time t its velocity is v , then

$$\text{acceleration (a)} = \frac{v - u}{t}$$

Its SI unit is m/sec^2 or $\text{m}.\text{sec}^{-2}$.

Newton's Laws of Motion :

- Newton's laws of motion are three physical laws that, together, laid the foundation for classical mechanics. They describe the relationship between a body and the forces acting upon it and its motion in response to those forces.

1. Newton's First Law of Motion –

- Newton's first law states that every object will remain at rest or in uniform motion in a straight line unless compelled to change its state by the action of an external force.
- Inertia is a tendency of matter by which it continues in its existing state of rest or uniform motion in a straight line unless that state is changed by an external force.
- The tendency of an object to resist changes in its state of motion varies with mass. Mass is that quantity which is solely dependent upon the inertia of an object. The more inertia that an object has the more mass that it has. The light object has less inertia.

2. Newton's Second Law of Motion –

- The second law states that the acceleration of an object depends upon two variables - the net force acting upon the object and the mass of the object.
- Thus, the acceleration of an object subject to a force is directly proportional to the net force, in the same direction as the net force, and inversely proportional to the mass of the object i.e. $a = F/m$

Where a is acceleration, F = force & m = mass

or $F = m \times a$ $\square$ Force = Mass $\times$ Acceleration

SI unit of force is newton.

- 1 newton is equal to 1 kilogram metre per second square or 1 newton of force is the force required to accelerate an object with a mass of 1 kilogram, 1 metre per second, every second.

3. Newton's Third Law of Motion –

- For every action, there is an equal and opposite reaction.
- The Force imposed by the first object to other object is known as an action while the force imposed by the second object on the first object is known as reaction. The size of the force on the first object equals the size of the force on second object. Newton's third law of motion is also known as 'Action-Reaction' Law.

Momentum :

- Momentum is the product of the mass & velocity of an object.

$$\backslash \quad \text{Momentum (P)} = \text{Mass (m)} \times \text{Velocity (v)}$$

- It is a vector quantity possessing a magnitude and a direction. Its SI unit is kilogram metre per second ($\text{kg}.\text{m/sec}.$)

Change in momentum –

- The rate of change in momentum of an object is proportional to the imposed force on an object.
- The change in momentum occurs always in the direction of the force. It is the other form of newton's second law of motion.

$$\text{Force} \propto \frac{\text{change in momentum}}{\text{time}}$$

If an object of m mass has its initial velocity u , then its initial momentum = mu

Its last velocity became v , after the imposing of force (F) for time (t). So, the final momentum of the object = mv

$$\backslash \quad \text{change in momentum} = mv - mu$$

$$\text{So, } F \propto \frac{mv - mu}{t}$$

$$\square \quad F \propto \frac{m(v - u)}{t}$$

$$\square \quad F \propto ma$$

$$\text{So, } F = Kma$$

In SI units, value of constant K is 1

$$\backslash \quad F = 1ma$$

$$\square \quad F = ma$$

It is the Newton's second law of motion.

Work :

- A force is said to do work if, when acting, there is a displacement of the point of application in the direction of force. If no displacement occurs after applying force, no work is done.

The value of work is equal to -

work = Force $\times$ displacement in the direction of force

or $W = F \times S$ where, $W = \text{work}$

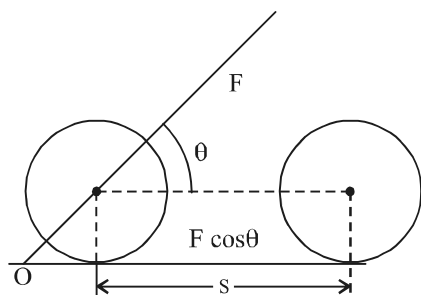
$F = \text{force}$

$S = \text{displacement}$

If the force (F) is not in direction of displacement, forming θ angle with it, then

work done (W) = $F \cos \theta \times S$

where, $F \cos \theta$ is the component of force in the direction of displacement



- Work is a scalar quantity. Its SI unit is joule. One joule is defined as the amount of energy exerted when a force of 1 newton is applied over a displacement of one metre.
1 Joule = 1 newton $\times$ 1 metre.

Power :

- Power is the rate of doing work by machine or anybody.

$$\text{Power} = \frac{\text{work (w)}}{\text{time (t)}}$$

- Its SI unit is joule/sec. It is also called watt.
- Another unit of Power is Horse Power.
1 Horse Power = 746 watt.

Energy :

- Energy is the capacity for doing work. Different forms of energy are Mechanical energy, Heat energy, Light energy, Sound energy, etc.
- It is a scalar quantity. Its SI Unit is joule.

Mechanical Energy :

- Mechanical energy is the sum of kinetic & potential energy in an object that is used to do work. The sum of potential and kinetic energy is called mechanical energy.
- So, mechanical energy = kinetic energy + potential energy

Kinetic Energy :

- The kinetic energy of an object is the energy that it possesses due to its motion. If the mass of the body is m kg and speed v m/sec. then

$$\text{K.E.} = \frac{1}{2}mv^2$$

Potential Energy :

- Potential energy is the energy in a body due to its position. It is clear that if any object under a force is shifted from one position to another position, then during it, done work is stored as the potential energy of the object.

P.E. = mgh where, $m = \text{mass of the object}$

$g = \text{gravitational acceleration}$

$h = \text{height}$

- Elastic potential energy, gravitational potential energy, electrical potential energy etc. are different forms of potential energy.

Transformation of Energy by Apparatus

S.No.	Name of Apparatus	Transformation of Energy
1.	Dynamo	Conversion of mechanical energy into electrical energy.
2.	Electric motor	Conversion of electrical energy into mechanical energy.
3.	Microphone	Conversion of sound energy into electrical energy.
4.	Loudspeaker	Conversion of electric energy into sound energy.
5.	Solar cell	Conversion of solar energy into electrical energy.
6.	Tubelight	Conversion of electric energy into light energy.
7.	Electric bulb	Conversion of electric energy into heat energy and light energy.
8.	Electric cell	Conversion of chemical energy into electric energy.
9.	Candle	Conversion of chemical energy into light & heat energy.
10.	Sitar	Conversion of mechanical energy into sound energy.
Note : During photosynthesis in green plants, solar energy is converted into chemical energy.		

Question Bank

1. A simple machine helps a person in doing :

- (a) Less work
- (b) The same amount of work with lesser force
- (c) The same amount of work slowly
- (d) The same amount of work much faster

I.A.S. (Pre) 1996

Ans. (b)

A simple machine helps a person in doing the same amount of work with less force. Few examples of simple machines are pulley, lever, wheel, screw etc.

2. A man is standing on a sensitive balance. If he inhales deeply, the reading of the balance –

- (a) Increases
- (b) Decreases
- (c) Remains unaffected
- (d) May increase or decrease depending on the atmospheric pressure

U.P.P.C.S. (Mains) 2013

Ans. (a)

The reading of the balance increases if the person inhales deeply. One litre of air weighs about 1.225 g. A maximal 4.8 litre breath (average vital capacity for males) weighs approximately 5.88g. Inhaled air will add to the mass of the man and thus his weight goes up slightly.

3. Which of the following is not a vector quantity?

- (a) Displacement
- (b) Velocity
- (c) Force
- (d) Volume

45th B.P.S.C. (Pre) 2001

Ans. (d)

The scalar quantity is defined as the quantity that has only got magnitude, but not directions whereas, vector quantity comprises of both magnitude and directions. Displacement, velocity and force are the examples of vector quantity while volume is a scalar quantity.

4. Which of the following is a vector quantity ?

- (a) Time
- (b) Speed
- (c) Displacement
- (d) Distance

U.P.R.O./A.R.O. (Mains) 2014

Ans. (c)

See the explanation of above question.

5. Which of the following is not a vector quantity?

- (a) Speed
- (b) Velocity
- (c) Torque
- (d) Displacement
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Speed is a scalar quantity while velocity, torque and displacement are the examples of vector quantity.

6. Which one of the following is a scalar quantity?

- (a) Force
- (b) Pressure
- (c) Velocity
- (d) Acceleration
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Force, velocity and acceleration are the examples of vector quantity while pressure is a scalar quantity. Pressure is defined as the force per unit area. It is the force acting in all direction on an area while force is a pull or push in a specific direction.

7. Which one of the following is a vector quantity ?

- (a) Momentum
- (b) Pressure
- (c) Energy
- (d) Work

I.A.S. (Pre) 1997

Ans. (a)

A vector quantity has both magnitude and direction, e.g. velocity, displacement, acceleration, momentum, force, etc. A scalar quantity has only magnitude, e.g. distance, volume, speed, pressure, energy, work, density, etc.

8. Which physical quantity is represented by the ratio of momentum and velocity of the body?

- (a) Velocity
- (b) Acceleration
- (c) Mass
- (d) Force

45th B.P.S.C. (Pre) 2001

Ans. (c)

Momentum is a vector quantity that is the product of mass and the velocity of an object or particle.

Momentum = Mass × Velocity

So, $\text{Mass} = \frac{\text{Momentum}}{\text{Velocity}}$

9. Force is the product of –

- (a) Mass and velocity
- (b) Mass and acceleration
- (c) Weight and velocity
- (d) Weight and acceleration

45th B.P.S.C. (Pre) 2001

Ans. (b)

A force can cause an object with mass to change its velocity (which includes to start moving from a state of rest) i.e., to accelerate.

Force (F) = Mass × Acceleration = m.a

10. Which of the following is a non-contact force ?

- (a) Magnetic force
- (b) Frictional force
- (c) Impact force
- (d) None of these

70th B.P.S.C. (Pre) 2024

Ans. (a)

Non-contact forces act on objects without the need for physical contact. For example : Gravitational force – The force of attraction between any two objects with mass, causing objects to fall towards the Earth. Magnetic force – The force exerted by magnets, causing them to attract or repel to each other. Electric force – The force exerted by electric charges, causing them to attract or repel each other. On the other hand, contact forces require physical contact between objects for the force to act. Frictional force and impact or applied force are examples of contact force.

11. If the number representing volume and surface area of a cube are equal, then the length of the edge of the cube in terms of the unit of measurement will be –

- (a) 3 (b) 4
(c) 5 (d) 6

I.A.S. (Pre) 1997

Ans. (d)

Volume of cube = l^3 (where l is the length of cube's edge)
Surface area of cube = $6l^2$
According to question, both are equal to each other
Therefore $l^3 = 6l^2$
Hence $l = 6$

12. Energy conservation means –

- (a) Generation and destruction of energy
(b) Energy could be created
(c) Energy could not be created but destroyed
(d) Energy can neither be created nor destroyed

M.P.P.C.S. (Pre) 2000

Ans. (d)

The Law of conservation of energy states that energy can neither be created nor destroyed; rather, it can only be transformed or transferred from one form to another.

13. Most of the sources of energy we use, represent stored solar energy. Which of the following is NOT ultimately derived from the Sun's energy?

- (a) Nuclear energy (b) Geothermal energy
(c) Biomass energy (d) Wind energy

U.P. P.C.S. (Pre) 2022

Ans. (a) & (b)

Geothermal energy is heat that is generated within the Earth. It comes from heat generated during the original formation of the planet and the radioactive decay of materials. This thermal energy is stored in rocks and fluids in the centre of the earth. Nuclear energy is a form of energy released from the nucleus, the core of atoms, made up of protons and neutrons. This source of energy can be produced in two ways : fission – when nuclei of atoms split into several parts; or fusion – when nuclei fuse together. Hence, geothermal energy

and nuclear energy both are not derived from the Sun's energy. Biomass energy and Wind energy are ultimately derived from the Sun's energy.

14. The energy of wind is –

- (a) Only potential (b) Only kinetic
(c) Electrical (d) Potential and kinetic both

Uttarakhand Lower Sub. (Pre) 2010

Ans. (b)

All moving things have kinetic energy. It is energy possessed by an object due to its motion or movement. Wind energy is the kinetic energy of air in motion. The kinetic energy of the wind (wind energy) can be converted into mechanical and electrical energy.

15. In wind power, which form of energy is converted into electrical energy?

- (a) Kinetic energy (b) Potential energy
(c) Solar energy (d) Radiant energy

U.P. PCS (Pre) 2016

Ans. (a)

The term wind energy or wind power describes the process by which the wind is used to generate mechanical power or electricity. Wind turbines convert the kinetic energy of the wind into mechanical power. This mechanical power can be used for specific tasks such as grinding grain or pumping water or a generator can convert this mechanical power into electricity.

16. Which of the following energy changes involves frictional force?

- (a) Kinetic energy to heat energy
(b) Potential energy to sound energy
(c) Chemical energy to heat energy
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Frictional forces apply in a static condition as well as in a moving condition. The respective co-efficient of friction are called static co-efficient and dynamic co-efficient. While in the static condition, there is no displacement i.e. no work done and hence no energy conversion. It is in the dynamic condition, there is movement against the frictional force that there is energy loss by way of heat. In other words, kinetic energy of movement is converted into heat.

Energy conversion due to friction does not involve change in potential energy/chemical energy. Even when an object is falling in air, its potential energy first gets converted into kinetic energy and then heat or sound etc. In battery, its chemical energy is converted into electrical energy as well as heat but friction has nothing to do in such conversion.

17. Match List-I with List-II and choose the correct answer from the code given below-

List-I (Energy Conversion)	List-II (Device/Mechanism)
A. Light to electric	1. Car Braking
B. Electric to sound	2. Nuclear reactor
C. Mass to heat	3. Loudspeaker
D. Chemical to heat and light	4. Solar cell
	5. Fuel combustion

Code :

	A	B	C	D
(a)	1	2	3	4
(b)	4	3	2	5
(c)	2	1	3	5
(d)	3	1	2	4

U.P.P.C.S. (Mains) 2002

Ans. (b)

Energy conversion	Device/mechanism
Light to electric	– Solar cell
Electric to sound	– Loudspeaker
Mass to heat	– Nuclear reactor
Chemical to heat and light	– Fuel combustion

18. Who is the first person to define speed?

- (a) Galileo (b) Newton
(c) Kepler (d) Ptolemy
(e) None of the above/More than one of the above

67th B.P.S.C. (Pre) (Re. Exam) 2022

Ans. (a)

Italian physicist Galileo Galilei is usually credited with being the first to measure speed by considering the distance covered and the time it takes. Galileo defined speed as the distance covered per unit of time. In equation form, that is $v = d/t$ (where v is speed, d is distance, and t is time). After that Newton propounded the laws of motion to describe the motion of bodies and objects.

19. Newton's first law of motion is known as :

- (a) Gravity
(b) Law of Inertia
(c) Law of conservation of momentum
(d) Vertical motion

Chhattisgarh P.C.S. (Pre) 2019

Ans. (b)

Newton's first law of motion is also known as 'Law of Inertia'. It states that, if an object is at rest or moving at a constant speed in a straight line, it will remain at rest or keep moving in a straight line at constant speed unless acted upon by an unbalanced force. Inertia is the tendency of an object to remain in a state of rest or uniform motion unless a force acts upon it.

20. As the train starts moving, the head of the passenger sitting inside leans backward because of –

- (a) Inertia of Rest (b) Inertia of Motion
(c) Moment of Inertia (d) Conservation of Mass

U.P. U.D.A./L.D.A. (Pre) 2010

Ans. (a)

According to the Newton's first law of motion sometimes referred to as the law of inertia which states that an object at rest stays at rest and in motion remain in motion with same speed and in the same direction unless acted upon by an external force. When a train suddenly starts, the passenger tends to fall backward. This is because the lower part of the body which is in contact with the train begins to move while the upper part of the body tends to maintain its position due to inertia of rest. As a result, the upper part tends to fall backward.

21. A bus is moving along a straight path and takes a sharp turn to the right side suddenly. The passengers sitting in the bus will :

- (a) bent towards right side
(b) fall in the forward direction
(c) bent towards left side
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (c)

If a bus is moving along a straight path and takes a sharp turn to the right side suddenly, then the passengers sitting on the bus will bend towards the left side. Initially, the passengers and the bus both are moving in the same direction. When the bus suddenly takes a right turn, the lower portion of passengers who are in contact with the bus tends to move in that direction but the upper portion of the passengers will be thrown towards the left. This is because of inertia of direction.

22. Static science is associated with –

- (a) Dynamic situation
(b) Situation of rest
(c) Mental situation
(d) Data analyzing

M.P. P.C.S. (Pre) 2012

Ans. (b)

Statics is a branch of mechanics associated with the situation of rest.

23. Assertion (A) : A man standing on a completely frictionless surface can propel himself by whistling.

Reason (R) : If no external force acts on a system, its momentum cannot change.

Code :

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not a correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

I.A.S. (Pre) 2000

Ans. (b)

First statement can be explained on the basis of Newton's third law of motion, "For every action, there is equal and opposite reaction." While whistling, a man uses his breath as a propellant to propel himself. Momentum is the property that a moving object has due to its mass and motion. Hence, when there is no external force applied on any moving object, its momentum cannot change. Hence, statement A and R both are correct but R is not a correct explanation of A.

24. It is difficult to walk on the ice than on the road because :

- (a) Ice is harder than the road.
- (b) Road is harder than the ice.
- (c) Ice does not offer any reaction when we push it with our foot.
- (d) Ice has a lesser friction than the road.

U.P.P.C.S. (Pre) 1994

Ans. (d)

It is very difficult to walk on the ice than on the road because ice has a lesser friction than the road. The roughness of road's surface gives you much frictional force which enables you to move forward on road.

25. Consider the following statement and conclusions which can be drawn from it. Choose the correct conclusion.

Statement : In starting a loaded cart, one has to push harder than to keep it moving.

Conclusion :

- (a) The weight of a moving object is less.
- (b) The wheels tend to slip initially.
- (c) There is less friction once the cart starts moving.
- (d) Practice makes perfect.

U.P.P.C.S. (Mains) 2013

Ans. (c)

The static friction that holds an object in place is greater than the kinetic friction that slows down a moving object. In other words, once you start an object moving, the friction decreases from the static friction holding the object in place. You have seen this in trying to slide a heavy box across the floor. It may be very difficult to move, but once it starts sliding, it is easier to push.

26. A goalkeeper in a game of football pulls his hands backwards after holding the ball shot at the goal. This enables the goalkeeper to :

- (a) decrease the rate of change of momentum
- (b) exert large force on the ball
- (c) increase the force exerted by the ball on hands
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

According to Newton's second law,

$$F = m \times a$$

$$\text{Or, } F = m (v - u)/t$$

So, by pulling his hands backwards, the goalkeeper increases time to hold the ball. By increasing the time, he decreases the force (rate of change of momentum) applied by the football on him. In other words, the goalkeeper pulls his hands backward while holding the ball shot at the goal to reduce the rate of change of momentum which in turn reduces the force exerted on the hands by the ball.

27. For every action there is an equal and opposite reaction.

The law was given by :

- (a) Newton
- (b) Albert Einstein
- (c) Bohr
- (d) Handane

U.P. R.O./A.R.O. (Mains) 2017

Ans. (a)

For every action, there is an equal and opposite reaction is known as Newton's third law of motion. The statement means that in every interaction, there is a pair of forces acting on the two interacting objects. The size of the force on the first object equals the size of the force on the second object. The direction of the force on the first object is opposite to the direction of the force on the second object.

28. Centripetal force is responsible to :

- (a) independent motion of the object in space
- (b) keep the body moving along the circular path
- (c) fly the object along a straight line
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (b)

Newton's first law of motion states that an object moving at constant speed will continue that motion unless acted on by an outside force. This means that circular motion can only happen if there is a 'centre seeking' force – otherwise things would just travel in a straight line, rather than the curved line of a circle. Centripetal means 'centre seeking', so centripetal force is used to refer to the force experienced by an object travelling in a circle. For example, when someone spins a

ball attached to a rope horizontally above his head, the rope transmits a centripetal force from the muscles of the hand and arm, causing the ball to move in a circular path. In the special case of the Earth's circular motion around the Sun – or any satellite's circular motion around any celestial body – the centripetal force causing the motion is the result of the gravitational attraction between them.

29. An effective Coriolis force results from :

- (a) Solar system
- (b) Earth rotation
- (c) interior of the Earth
- (d) Colorado and Gulf Streams
- (e) None of the above/More than one of the above

67th B.P.S.C. (Pre) 2022

Ans. (b)

In physics, the Coriolis force is an inertial or fictitious force that acts on objects in motion within a frame of reference that rotates with respect to an inertial frame. In popular (non-technical) usage of the term 'Coriolis effect', the rotating reference frame implied is almost always the Earth. An effective Coriolis force is caused due to rotation of the Earth. This force causes moving objects on the surface of the Earth to be deflected to the right (with respect to the direction of travel) in the Northern Hemisphere and to the left in the Southern Hemisphere.

30. A man jumped at a speed of 5 metre per second from a stationary boat and the boat moved off with the speed of 0.5 metre per second. How many times is the mass of the boat greater than that of the man?

- (a) 5.5 times
- (b) 4.5 times
- (c) 2.5 times
- (d) 10 times

I.A.S. (Pre) 1994

Ans. (d)

Let the mass of the person is M and the mass of the boat is N , then momentum of boat + momentum of person before Jump = 0

then Momentum of both after man's jump = $M \times 5 - N \times 0.5$

$$\square M \times 5 - N \times 0.5 = 0 \quad \left(\text{According to the principle of conservation of momentum} \right)$$

$$\backslash \quad N \times 0.5 = M \times 5 \quad \square \quad \frac{N}{M} = \frac{5}{0.5} = 10$$

Thus mass of the boat is 10 times greater than that of the man.

31. What will be the de Broglie wavelength of a ball of mass 0.12 kg moving with a speed of 20 ms⁻¹?

- (a) 6.63×10^{-30} m
- (b) 6.63×10^{-34} m

(c) 2.76×10^{-30} m

(d) 2.76×10^{-34} m

Uttarakhand P.C.S. (Pre) 2024

Ans. (d)

The wavelength (λ) that is associated with an object in relation to its momentum and mass is known as the de Broglie wavelength.

$$\text{de Broglie wavelength } \lambda = \frac{h}{p}$$

where, h = Planck constant

$$= 6.63 \times 10^{-34} \text{ joule-sec.}$$

$$p = \text{momentum} = \text{mass} \times \text{velocity}$$

$$= m.v$$

$$\text{Given that, } m = 0.12 \text{ kg}$$

$$v = 20 \text{ m/sec.}$$

\ de Broglie wavelength of the ball

$$\lambda = \frac{h}{p} = \frac{h}{m.v} = \frac{6.63 \times 10^{-34}}{0.12 \times 20} \text{ m} \\ = 2.76 \times 10^{-34} \text{ m}$$

Hence, option (d) is the correct answer.

32. A truck, a car, and a motorcycle have equal kinetic energies. If equal stopping forces are applied and they stop after traveling a distance of X, Y and Z respectively, then –

- (a) $X > Y > Z$
- (b) $X < Y < Z$
- (c) $X = Y = Z$
- (d) $X \cong 4Y \cong 8Z$

I.A.S. (Pre) 1996

Ans. (c)

If the objects have equal kinetic energy and if equal stopping force is applied on them, then they would stop at the same distance. Hence $X = Y = Z$.

33. The ratio of kinetic energies of two bodies of same mass is 4 : 9, the ratio of their velocities will be :

- (a) 4 : 9
- (b) 2 : 3
- (c) 16 : 81
- (d) $\sqrt{2} : \sqrt{3}$

U.P. R.O./A.R.O. (Pre) 2017

Ans. (b)

As per question mass of both bodies is same. Suppose, velocity of the body I is v_1 and Velocity of the body II is v_2 .

$$\text{Kinetic Energy of the body I } (K_1) = \frac{1}{2} m v_1^2$$

$$\text{Kinetic Energy of the body II } (K_2) = \frac{1}{2} m v_2^2$$

$$\frac{K_1}{K_2} = \frac{1/2 m v_1^2}{1/2 m v_2^2} = \frac{v_1^2}{v_2^2}$$

According to question,

$$\frac{K_1}{K_2} = \frac{4}{9} \Rightarrow \frac{v_1^2}{v_2^2} = \frac{4}{9} \Rightarrow \left(\frac{v_1}{v_2} \right)^2 = \frac{4}{9}$$

$$\Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{4}{9}} = \frac{2}{3} = 2 : 3$$

34. Cycle and cars used ball bearing, because –

- (a) The actual area of contact between the wheel and axle is increased.
- (b) The effective area of contact between the wheel and axle is increased.
- (c) The effective area of contact between the wheel and axle is reduced.
- (d) None of the above statement is correct.

I.A.S. (Pre) 2013

Ans. (c)

The purpose of a ball bearing is to reduce rotational friction and support radial and axial loads. Due to ball bearing, the effective area of contact between the wheel and axle is reduced and hence, the frictional force also reduced.

35. Ball bearings are used to convert static friction into:

- (a) rolling friction
- (b) drag
- (c) sliding friction
- (d) More than one of the above
- (e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Ball bearings are types of rolling-elements bearings that carry loads, reduce friction, and position moving machine parts while facilitating motion. The purpose of a ball bearing is to reduce rotational friction and support radial and axial loads. It achieves this by using at least two races to contain the balls and transmit the loads through the balls. In most applications, one race is stationary and the other is attached to the rotating assembly (e.g., a hub or shaft). As one of the bearing races rotates it causes the balls to rotate as well. Because the balls are rolling they have a much lower coefficient of friction than if two flat surfaces were sliding against each other. They reduce surface contact and friction across moving planes. Thus, ball bearings are used to convert static friction into rolling friction.

36. A person is sitting in a car which is at rest. The reaction from the road at each of the four wheels of the car is R. When the car runs on a straight level road, how will the reaction at either of the front wheels vary?

- (a) It will be greater than R
- (b) It will be less than R
- (c) It will be equal to R
- (d) It shall depend on the material of the road

I.A.S. (Pre) 2008

Ans. (c)

Car is in vertical equilibrium in case of rest or in horizontal motion and its weight should be equally distributed on all the wheels that creates equal reaction on wheels (in both cases) from the road, which maintains the car in equilibrium, otherwise torque will produce and car will topple. Hence, option (c) is the correct answer.

37. An oil tanker is partially filled with oil and moves forward on a level road with uniform acceleration. The free surface of oil then –

- (a) Remains horizontal
- (b) Is inclined to the horizontal with smaller depth at the rear end
- (c) Is inclined to the horizontal with larger depth at the rear end
- (d) Assumes parabolic curve

I.A.S. (Pre) 2003

Ans. (c)

When an oil tanker is partially filled with oil and moves forward on a level road with uniform acceleration, the free surface of oil then is inclined to the horizontal with larger depth at the rear end.

38. Consider the following statements :

A 4-wheel vehicle moving in a sharp circular path at high speed will :

1. Overturn about its outer wheels
2. Overturn about its inner wheels
3. Skid outwards
4. Skid inwards

Which of these statements are correct ?

- (a) 1 and 3
- (b) 2 and 4
- (c) 2 and 3
- (d) 1 and 4

I.A.S. (Pre) 2003

Ans. (a)

In case of object moving along a circular path two types of forces centripetal and centrifugal acts upon it. As they have opposite impact thus the object remains in balance. If a four wheeler on a circular path is moving faster than the relative speed then it skids outwards but then the frictional force provides the necessary centripetal force which ensures the car to have a curved path. On sharp circular path, if the outer portion of the road is not raised, then the vehicle with high speed will skid outwards and will overturn about its outer wheels.

39. A batsman hits a cricket ball which then rolls on a level ground. After covering a short distance, the ball comes to rest. The ball slows to a stop because

- (a) The batsman did not hit the ball hard enough
- (b) Velocity is proportional to the force exerted on the ball
- (c) There is a force on the ball opposing the motion
- (d) There is no unbalanced force on the ball, so the ball would try to come to rest

M.P.P.C.S. (Pre) 2017

Ans. (c)

A batsman hits a cricket ball which then rolls on a level ground. After covering a short distance, the ball comes to rest because there is an external force i.e. frictional force acting on the moving ball opposing its motion. If there is no frictional force then the ball will roll continuously.